Assignment – 1.8

Configure IPv6 Addressing

Q - 1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

- **Assign Static IPv6 Address:**
- **For Windows:**

1. Open Command Prompt as Administrator.
2. Type:

   ipconfig
3. Set the IPv6 address:

   netsh interface ipv6 set address interface="Ethernet" address=fd12:3456:789a::10
4. Verify the address:

   ipconfig

- **For Linux:**

1. Open the Terminal.

2. Assign the IPv6 address:

   sudo ip -6 addr add fd12:3456:789a::10/64 dev eth0

3. Verify the address:

   ip -6 addr show

- **Result:** The static IPv6 address is assigned and verified successfully.

Q - 2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Hint: Use the ping6 or ping -6 command for IPv6 testing.

- **Configure a Unique Local IPv6 Address:**
  1. Open **Command Prompt as Administrator**.
  2. Assign an IPv6 address:

     netsh interface ipv6 set address "Ethernet" fd00::10
  3. Check the address:

     ipconfig
  4. Ping another device on the same LAN:

     ping -6 fd00::20
- **Result:** If replies are received, the IPv6 connection is successful.

Q - 3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Constraint: Document the steps and the observed address behavior in a text file.

- **Temporary IPv6 Address:**
- **Steps:**
  1. Open Command Prompt or PowerShell as Administrator.
  2. Type ipconfig to check your IPv6 address.
  3. Disconnect and reconnect your network.
  4. Check the IPv6 address again.
  5. Restart your computer and check again.
  6. Note whether the address changes.
- **Result:** The temporary IPv6 address may change after reconnecting the network or restarting the computer.

Q - 4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

- **Python Program: Zomato-Style Restaurant Search:**

```
restaurants = {
    "Pizza Hut": "fd00::1",
    "Dominos": "fd00::2",
    "KFC": "fd00::3"
}
name = input("Enter restaurant name: ")
for r in restaurants:
    if name.lower() in r.lower():
        print(r, restaurants[r])
```

- **Output:**

```
Enter restaurant name: Pizza
Pizza Hut fd00::1
```

- **Result:** The program searches for a partial restaurant name and displays the matching restaurant's IPv6 address.

Q - 5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

➢ IPv6 Subnetting:

➢ AI-Generated Scenario:

- A company has one IPv6 network and needs 3 subnets.

Department	IPv6 Subnet
HR	2001:db8:abcd:1::/64
IT	2001:db8:abcd:2::/64
Sales	2001:db8:abcd:3::/64

➢ Configuration Steps (Windows):

1. Open Command Prompt as Administrator.
2. Check the network adapter name:

   netsh interface ipv6 show interfaces

3. Assign an IPv6 address:

   netsh interface ipv6 add address "Wi-Fi" 2001:db8:abcd:1::10

4. Check the address:

   ipconfig

5. Save the scenario and steps in Notepad as

   IPv6_Subnetting.txt.

- **Note:** Replace Wi-Fi with your actual adapter name. The example address is for documentation only and will not provide normal internet connectivity. Use an IPv6 prefix suitable for your network for real connectivity.